Quadratics: Expanding & Factorising

Practice test 2 · 40 marks · Higher tier · non-calculator

Name: _____ Date: _____ Score: _____ / 40

Section A — Expanding double brackets

A1. Expand and simplify $(x + 3)(x + 9)$

[2 marks]

A2. Expand and simplify $(x – 8)(x – 2)$

[2 marks]

A3. Expand and simplify $(2x + 5)^2$

[2 marks]

A4. A student writes $(x + 4)^2 = x^2 + 16$. Explain the mistake and write the correct expansion.

[2 marks]

Section B — Expanding three brackets

B1. Expand and simplify $(x + 3)(x – 1)(x – 4)$

[3 marks]

B2. Expand and simplify $(x + 2)^3$

[3 marks]

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Factorise $x^2 + 2x - 24$

[2 marks]

C2. Factorise $x^2 - 13x + 36$

[2 marks]

C3. Factorise $9x^2 - 16$

[1 mark]

C4. $x^2 + bx + 15$ factorises to $(x + 3)(x + a)$. Find the values of a and b.

[2 marks]

Section D — Factorising $ax^2 + bx + c$

D1. Factorise $5x^2 + 13x - 6$

[2 marks]

D2. Factorise $6x^2 - 11x + 4$

[2 marks]

D3. Factorise fully $3x^3 - 12x$

[2 marks]

Section E — Using factorising

E1. Solve $2x^2 + x - 15 = 0$

[3 marks]

E2. Simplify fully

$$\frac{2x^2 - 7x - 4}{x^2 - 16}$$

[3 marks]

Section F — Problem solving & proof

F1. (a) Write $x^2 + 8x + 5$ in the form $(x + a)^2 + b$. (b) Hence write down the coordinates of the turning point of $y = x^2 + 8x + 5$.

[3 marks]

F2. Prove that the difference between the squares of two consecutive odd numbers is always a multiple of 8.

[3 marks]

F3. Without a calculator, work out $1001^2 - 999^2$

[1 mark]